

Based on how they manage network traffic and guide data to its destination, here are the 3 key differences between a router and a switch:

1. Network Layer vs. Data Link Layer (IP vs. MAC Addresses)

- **Router:** Operates at **Layer 3 (Network Layer)** of the OSI model. It reads the source and destination **IP addresses** of incoming data packets to determine the best path across different networks.
- **Switch:** Operates at **Layer 2 (Data Link Layer)**. It ignores IP addresses entirely and looks at the physical **MAC (Media Access Control) addresses** embedded in data frames to pass information between devices on the same local network.

2. Connecting Different Networks vs. Creating a Local Network

- **Router:** Its main function is to **interconnect entirely distinct networks**. For example, it acts as the gateway linking your home's private Local Area Network (LAN) to the massive, public Wide Area Network (WAN) that is the internet.
- **Switch:** Its main function is to **create a network**. It provides the physical and logical ports needed to tie multiple local devices—like computers, printers, and servers—together into a single, shared LAN.

3. Data Routing Decisions vs. Data Forwarding Mechanisms

- **Router:** Uses a dynamically updated **Routing Table** to make complex traffic management decisions. It analyses network congestion, security protocols, and distance to "route" packets across boundaries, often changing the packet's path depending on network conditions.
- **Switch:** Uses a **MAC Address Table** (also called a CAM table) to map specific physical devices to specific physical ports. When data enters a port, the switch looks at its table and forwards the frame *only* to the exact port where the destination device is plugged in, keeping local traffic efficient and clear.